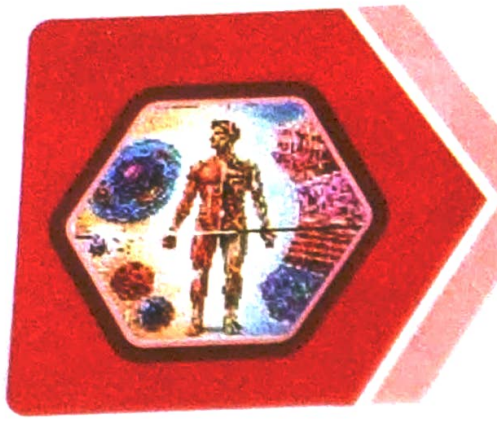


# INTRODUCTION TO HUMAN BODY & CELLULAR ADAPTATIONS

## LEARNING OBJECTIVES

**After studying this chapter, the learner will be able to:**

- Explain the basic concepts of anatomy, physiology, and pathophysiology.
- Describe the organization, systems, life processes, and homeostasis of the human body.
- Apply anatomical terms, planes, positions, and movements in anatomical descriptions.
- Explain the structure and functions of cells, the cell membrane, cytoplasm, nucleus, and cellular organelles.
- Classify the major tissue types and relate their structure to function.
- Explain the cell cycle, mitosis, and cytokinesis in growth and tissue maintenance.
- Relate common pathophysiological terms to cellular and tissue changes in disease.
- Describe cellular injury, oxidative stress, intracellular accumulations, and cellular adaptations.
- Differentiate the major types of cell death and necrosis.

## 1. INTRODUCTION TO HUMAN BODY

### 1.1. Definition and Scope of Anatomy, Physiology and Pathophysiology

❑ **Introduction of Anatomy**

Anatomy (ana = up; - tomy = process of cutting) is the science of body structures and the relationships among them.

❑ **Type of Anatomy**

1. **Macroscopic anatomy (Gross):** Body structures that can be observed without microscope.
2. **Microscopic anatomy:** Body structures that can be observed with microscope.

1. **Gross Anatomy**, also called macroscopic anatomy, is separated into 5 major divisions:

A. **Surface anatomy** describes surface forms and marks that can be understood by visualization and touch.

B. **Regional anatomy** describes the organization of specific areas of the body such as the head or hand. This approach is used mostly in professional schools: medical, dental, physical therapy.

C. **Systemic anatomy** describes groups of organs that function together for a single purpose.

D. **Developmental anatomy** describes the structural changes in an organism from fertilized egg to maturity. Embryology is the anatomical study of early development.

E. **Clinical anatomy** describes various medical specialties, including medical anatomy (changes that occur during illness), and radiographic anatomy.

2. **Microscopic Anatomy** is divided into two major divisions:

A. **Cytology,** the study of cells and their structures.

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B. **Histology**, the study of tissues and their structures.

## ❑ Introduction of Physiology

- Physiology (physio= nature -logy=study of) is a science that **deals with the study of functions of the body** and how body parts work together to maintain homeostasis.
- It includes the study of function of various organ systems like
  - ✓ **Cell Physiology**
  - ✓ **Tissue**
  - ✓ **The Blood**
  - ✓ **Digestive System**
  - ✓ **Respiratory System**
  - ✓ **Sense Organs**
  - ✓ **Skeletal System**
  - ✓ **Nervous System**
  - ✓ **Urinary System**
  - ✓ **Endocrine System**
  - ✓ **Reproductive System**
  - ✓ **Cardiovascular System**
  - ✓ **Lymphatic System**

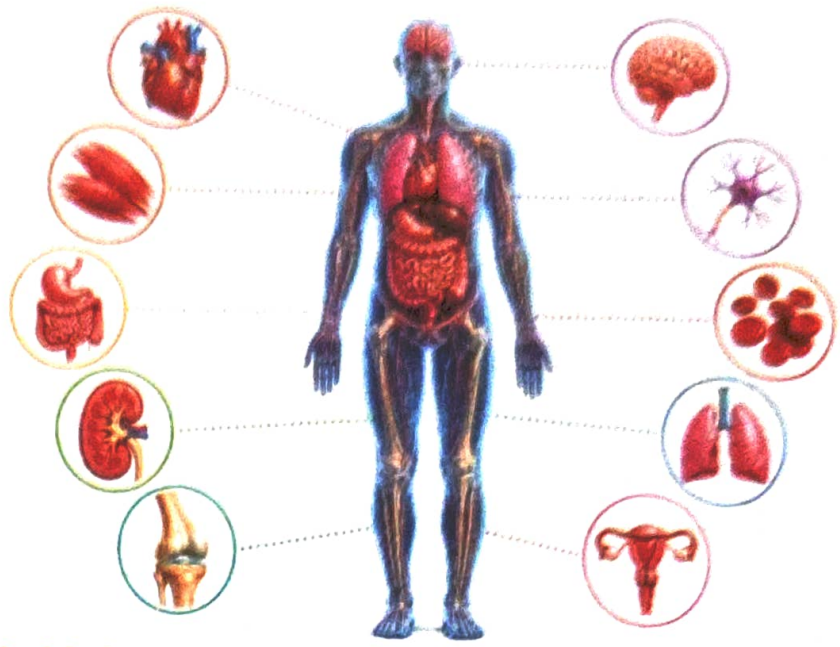


**Fig.1.1: Major Anatomical Systems of the Human Body**

**Table 1.1: Branches of Anatomy and Physiology**

| BRANCH | DEFINITION |
|---|---|
| Embryology | Study of the development of an organism from fertilization to birth |
| Developmental biology | Study of how organisms grow and develop at all stages of life |
| Cell biology | Study of cells, their structure, function, and processes |
| Histology | Study of tissues and their microscopic structure |
| Gross anatomy | Study of body structures visible to the naked eye |
| Systemic anatomy | Study of the body organized by organ systems (e.g., digestive, nervous system) |
| Cytology | Study of cells (structure, function, and abnormalities) |
| Nephrology | Study of kidneys |
| Pulmonology | Study of lungs and respiratory system |
| Myology | Study of muscles and their functions |
| Neurology | Study of the nervous system |
| Haematology | Study of blood |
| Endocrinology | Study of hormones and endocrine glands |
| Cardiology | Study of the heart |

## ❑ Introduction of Pathophysiology

- **Pathophysiology (or physiopathology)** is a branch of medical science that deals with the study of changes in body functions in a disease state. It is at the intersection of pathology and physiology.
- Rudolf Virchow coined the term **'Cellular pathology'.**
- Four components of Pathophysiology are
  - ✓ Etiology (Origin and cause of a disease)
  - ✓ Pathogenesis (Mechanism by which disease develops)
  - ✓ Morphology (Appearance of cell, tissue or organ)
  - ✓ Clinical features

Table 1.2: Important Terms of Pathophysiology

| TERMS | DEFINITION |
|---|---|
| Aplastic anemia | It is a condition that occurs when your body stops producing enough new blood cells |
| Chemotaxis | The movement of cells or an organism in response to a chemical stimulus |
| Hypoxia | Absence of enough oxygen in the tissue |
| Hepatitis | Inflammation of liver |
| Hypertrophy | Increase in cell size and tissue mass |
| Metaplasia | Change in cell type |
| Hypercortisolism | A condition that occurs from exposure to high cortisol level for long time |
| Neuroglia | They are non-neuronal cell in the CNS |
| Methemoglobinemia | It is a rare blood disorder where hemoglobin iron (ferrous) is oxidized to ferric state that cannot effectively release oxygen |
| Polycythemia | An abnormally increased in number of red blood cells in the blood |
| Osteoporosis | Condition in which bones become weak and brittle |
| Osteoclast | It is a type of bone cell that break down bone tissue |
| Osteoblast | Cell with single nucleus which form bones |
| Leukopenia | A low white blood cell count in blood |
| Aneurysm | An abnormal localized ballooning in the wall of a blood vessel (Arteries) |
| Apnea | A potentially serious sleep disorder in which breathing repeatedly stop and start |
| Amenorrhea | Absence of menstrual period |
| Acromegaly | Over production of Growth Hormone in adults |
| Agranulocytosis | When body doesn't make enough white blood cells (neutrophils, eosinophils and basophiles) |
| Apoptosis | A process of programmed cell death |
| Brucellosis | Bacterial infection spread from animal to humans |
| Bronchitis | Inflammation of the bronchi |
| Candidiasis | Fungal infection caused by candida |
| Cholestasis | A decrease or blockage in the flow of bile. |
| Chorio-carcinoma | Fast growing cancer that occur in women's uterus(womb) |
| Catalepsy | Condition characterized by lack of response to external stimuli and muscular rigidity. |
| Cyanosis | Bluish discoloration of the skin and mucous membrane due to lack of oxygen in blood |
| Dyspnea | Shortness of breath |
| Dysmenorrhea | Cramps and pelvic pain with menstruation |
| Dysplasia | Change in structure of each cell |
| Erythropoiesis | The process through which new red blood cells are created; it begins in the bone marrow. |
| Emphysema | Destruction of alveolar walls and permanent dilation of airspaces distal to terminal bronchioles (Shortness of breath) |
| Fibrosis | Excessive deposition of collagen fibers at the cell injury site. |
| Gestational hypertension | Pregnancy induced Hypertension |
| Granulocytopenia | A marked decrease in the number of granulocytes. |
| Hypocortisolism (Addison's Disease) | A disorder in which adrenal gland doesn't produce enough adrenocortical hormones. |
| Hypercortisolism (Cushing's syndrome) | A condition that occurs from exposure to high cortisol level for long time |

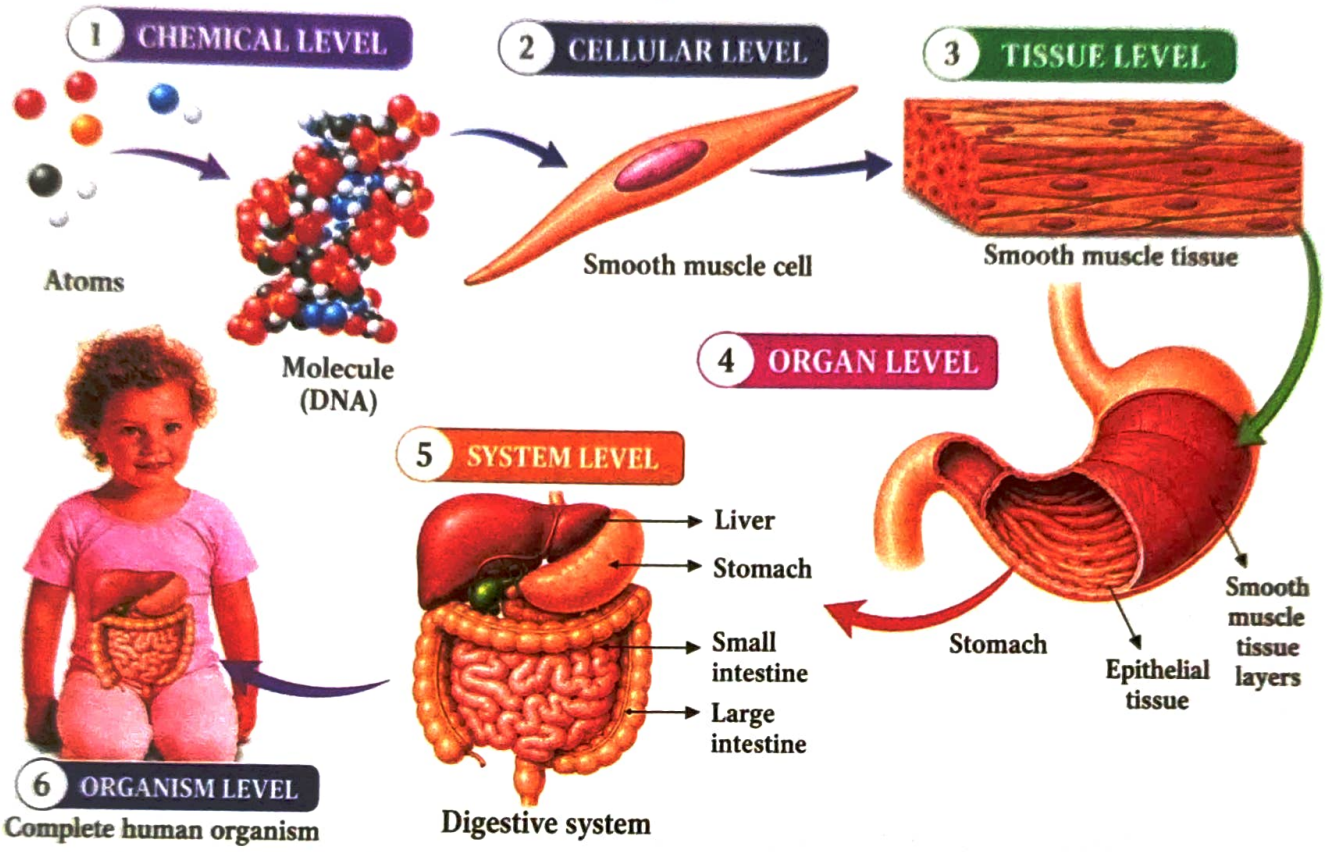

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| | |
|---|---|
| **Phagocytosis** | It is a cellular process for ingesting and eliminating particles (microorganisms, foreign substances) |
| **Pheochromocytoma** | A hormone secreting tumor that can occur in the adrenal gland |
| **Pyelonephritis** | Kidney infection due to bacteria |
| **Rhinorrhea** | Runny nose |

## 1.1.2. Levels of Structural Organization

### ❑ Introduction

- The human body is organized into six distinct levels of increasing complexity, from smallest to largest: **chemical, cellular, tissue, organ, organ system, and organism.**
- This level-wise structure ranges from **atoms** forming **molecules to tissues, organs, and systems** working together to sustain the entire **organism's life.**




**Atoms → Cells → Tissues → Organs → Systems → Body**

**Fig.1.2: Levels of Structural Organization**

### ❑ Six Levels of Organization

#### 1. Chemical Level

- The most fundamental level consists of **atoms, such as carbon, hydrogen, oxygen, nitrogen, phosphorus, and sulfur.**
- These atoms form **molecules and compounds** (such as **DNA, proteins, water** etc.), which are the building blocks of life.

#### 2. Cellular Level

- Molecules combine to form **cells, the basic functional units of life.** Cells (e.g., muscle cells, nerve cells) are the **smallest units of life,** capable of performing essential life processes.
- Their basic functions include growth, metabolism (chemical processes), response to stimuli (like irritability), and reproduction.

#### 3. Tissue Level

- Tissues are **groups of similar cells** working together to perform a specific function.
- The four primary tissue types are **epithelial, connective, muscle, and nervous.**

#### 4. Organ Level

- Where two or more distinct types of tissues combine to form a **structured unit, or organ**, that performs specific, complex functions.
- For example, the **stomach** includes all four tissue types to fulfill its role in digestion.

5. Organ System Level
   - An organ system is **a group of organs that work together** to achieve major physiological goals, such as the cardiovascular system (heart and blood vessels).
   - There are **11 major organ systems** in the human body. These are:
     1. **Integumentary system**
     2. **Skeletal system**
     3. **Muscular system**
     4. **Nervous system**
     5. **Endocrine system**
     6. **Cardiovascular system**
     7. **Lymphatic system**
     8. **Respiratory system**
     9. **Digestive system**
     10. **Urinary system**
     11. **Reproductive system**
6. Organismal Level
   - At the highest level of organization, **all systems function together to create an organism,** a complete living entity.
   - An organism is any living thing considered as a whole whether composed of one cell (such as bacteria) or trillions of cells (such as human).
   - The human organism is a complex of organ systems, all of which are mutually dependent on one another.

## 1.1.3. Body Systems

- Body systems are **groups of organs and tissues** that work together to carry out important and complex functions in the human body.
- For example, the circulatory system moves blood, the respiratory system helps in breathing, and the muscular system allows movement.
- These systems are interconnected and depend on each other. By working together, they help the body maintain a stable internal environment, which is called homeostasis.
- This stability includes maintaining proper temperature, oxygen levels, fluid balance, and other conditions necessary for survival.
- Some organs may be part of more than one body system if they serve more than one function.
- The human body consists of **11 organ systems**, each of which contains several functions and specific organs.

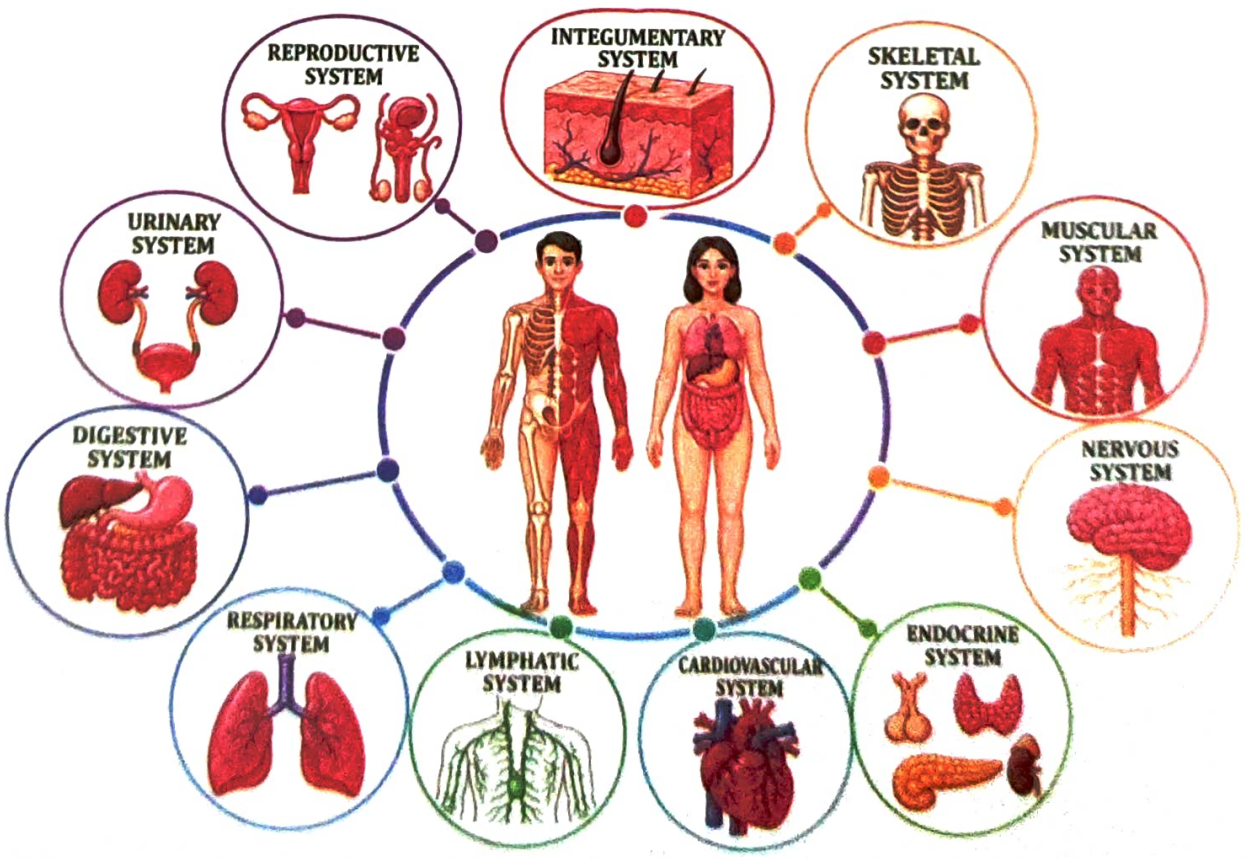



**Fig.1.3: Eleven Major Organ Systems of the Human Body**


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**Table 1.3: Body Systems and their Major Functions**

| SYSTEM | MAIN PARTS | FUNCTIONS |
|---|---|---|
| Integumentary System | Skin, Hair, Nails, Sweat Glands, Oil Glands | Protects the body from injury, infection, and harmful UV rays. It also helps in sensation, temperature regulation, and repair of damaged tissues. |
| Skeletal System | Bones, Cartilage, Joints, Ligaments, Bone Marrow | Provides shape, support, and posture to the body, protects vital organs, helps in movement, stores minerals like calcium, and produces blood cells in the bone marrow. |
| Muscular System | Skeletal Muscles, Smooth Muscles, Cardiac Muscles | Helps in movement of the body, maintains posture, produces heat, and moves substances such as food and blood within the body through contraction and relaxation. |
| Nervous System | Brain, Spinal Cord, Peripheral Nerves | Controls and coordinates body activities, transmits messages throughout the body, helps in sensation, and regulates voluntary and involuntary actions. |
| Endocrine System | Pituitary Gland, Thyroid Gland, Adrenal Glands, Pancreas, Ovaries, Testes | Secretes hormones directly into the blood to regulate metabolism, growth, reproduction, mood, and maintain homeostasis. |
| Cardiovascular System | Heart, Blood, Arteries, Veins, Capillaries | Circulates blood throughout the body to transport oxygen, nutrients, hormones, and remove waste products. |
| Lymphatic System | Lymph, Lymph Vessels, Lymph Nodes, Spleen, Thymus, Tonsils | Maintains fluid balance, absorbs fats from the intestine, and protects the body against infections. |
| Respiratory System | Nose, Pharynx, Larynx, Trachea, Bronchi, Lungs, Alveoli | Helps in breathing by taking in oxygen and removing carbon dioxide. It also helps in speech and maintaining pH balance. |
| Digestive System | Mouth, Tongue, Pharynx, Esophagus, Stomach, Small Intestine, Large Intestine, Rectum, Anus | Breaks down food into nutrients for energy, growth, and repair, absorbs nutrients, and eliminates waste from the body. |
| Urinary System | Kidneys, Ureters, Urinary Bladder, Urethra | Removes waste products such as urea and uric acid, maintains water and electrolyte balance, regulates blood pressure, and maintains blood pH. |
| Reproductive System | Testes, Penis, Ovaries, Uterus, Vagina | Produces sex cells and hormones, enables fertilization, supports fetal development, and helps in reproduction. |

## 1.1.4. Basic Life Processes

### ❑ Introduction

- All living organisms have certain features that make them different from non-living things.
- The basic life processes include organization (proper structure), metabolism (chemical reactions in the body), responsiveness (reacting to changes), movement, and reproduction (produce new individual).
- In humans, who are the most complex living beings, there are some extra processes like growth (increase in size), differentiation (cells becoming specialized), respiration (taking in oxygen and releasing energy), digestion (breaking down food), and excretion (removal of waste).
- All these processes are interconnected, and the body works as a coordinated system. Maintaining this balance is essential for health, as any disturbance can lead to diseases or even death. These processes together define the characteristics of human life.

### ❖ Organization

- At every level of the body, there is a proper division of work.
- Each part has its own specific function and works together with other parts.
- Even a single cell needs proper structure and coordination—if it loses its organization, it cannot survive.

### ❖ Metabolism

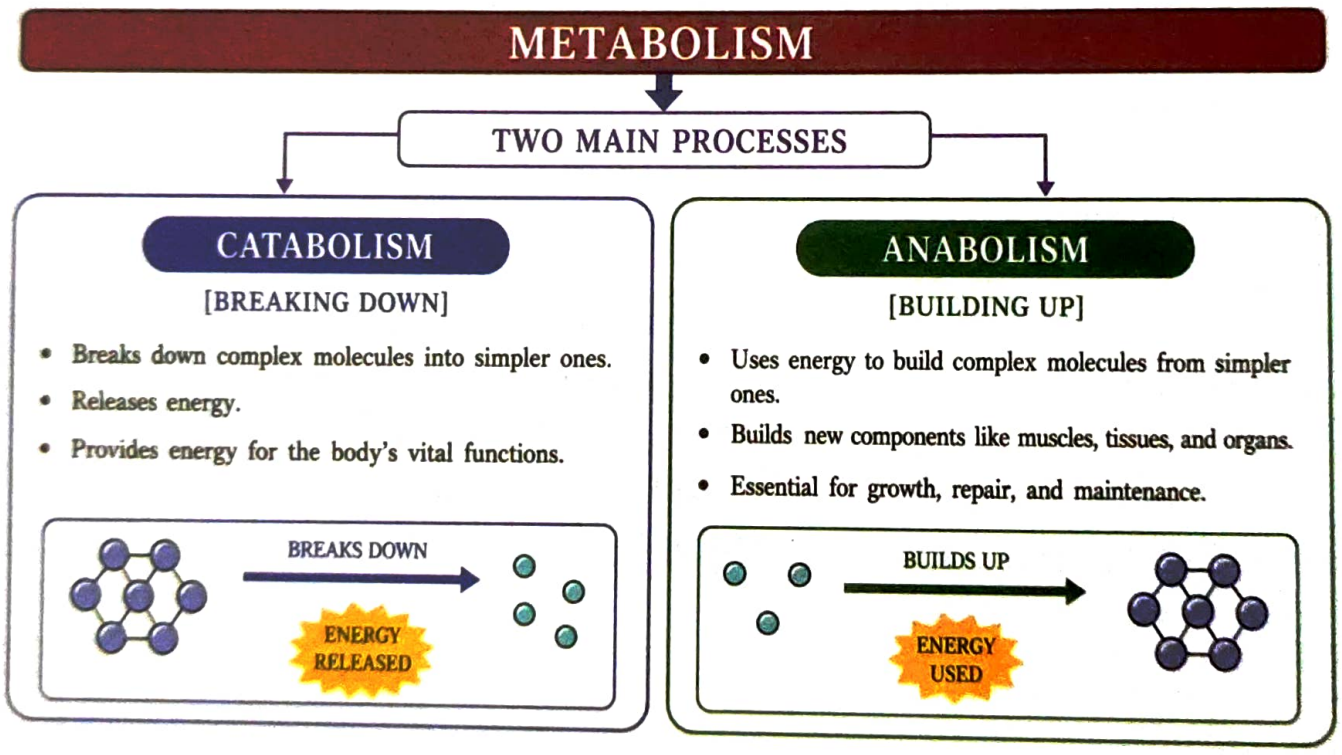
- Metabolism is a combination of series of reactions in the body to provide energy and synthesis of new organic material. These reactions help to control and maintain the living state of the cells in an organism.
- It is mainly divided into two types: **catabolism and anabolism**.

#### ➢ Catabolism

- Catabolism is the process in which **complex molecules** (proteins, lipids, nucleic acids, polysaccharides) **are broken down into smaller units** like amino acids, fatty acids, and monosaccharides.
- During this process, **energy is released**. It is called the destructive phase because it breaks down substances.
- This energy is used for cell activities, heat production, and making ATP (energy of the cell).

#### ➢ Anabolism

- Anabolism is the process of **building complex molecules from simpler ones**, like making proteins from amino acids or forming muscle tissue from nutrients.
- It **uses energy (ATP)**, so it is called an energy consuming process.
- This process helps in growth, repair of tissues, and maintaining body functions.




### ❖ Responsiveness

- Responsiveness is the ability of the body to detect changes inside or outside the environment and react to them. It means sensing a stimulus and giving an appropriate response.
- It helps in maintaining **homeostasis** and is important for survival.
- This response is often automatic and **allows the body to adjust** to things like **temperature changes, chemical signals, or danger.**

### ❖ Movement

- There are many types of movements in the body.
- At the cellular level, molecules move from one place to another.
- Blood flows throughout the body, and the diaphragm moves with every breath.
- Muscles help in movement by shortening, and this ability of muscle is called **contractility**.


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### ❖ Reproduction

- **Reproduction** is the basic biological process by which existing organisms (parents) produce new individuals, i.e., the birth of a baby. It ensures the **continuation of the species** and passes genetic information from parents to offspring.
- In a **wider sense**, reproduction also includes the formation of new cells in the body.
- These cells help in the **replacement of old cells, repair of tissues, and growth**—this is called **cellular reproduction**. **Both are essential** for the **continuity and survival of human life**.

### ❖ Growth

- All living things, including humans, grow and develop in the patterns determined by their DNA. Growth refers to an increase in size either through an increase in the number of cells or through an increase in the size of each individual cell.
- For growth to occur, **building processes (anabolic processes)** must happen faster than **breaking processes (catabolic processes)**.

### ❖ Differentiation

- Differentiation is a developmental process by which **unspecialized cells change into specialized cells** with distinct structural and functional characteristics.
- Through differentiation, cells develop into tissues and organs.For example, some cells become **skin cells**, while others become **bone cells, muscle cells, or other specialized cells**.

## 1.1.5. Homeostasis

### ❑ Introduction

- **Homeostasis** means maintaining a **stable internal environment** in the body (homeo = same, stasis = stable).
- This term was introduced by **Walter Bradford Cannon** in **1929**.
- The concept of internal environment was first explained by **Claude Bernard**, who called it **"milieu interieur"** (internal environment).
- He explained that humans and other organisms survive because their **internal environment is kept constant and well-controlled**, even if external conditions change.
- Homeostasis is the ability of the body to maintain a stable internal environment despite changes in external conditions.

### ❑ Internal Environment

**Internal environment** refers to the **extracellular fluid (ECF)** in which body cells exist.

- **ECF = fluid outside the cells**, and it continuously circulates throughout the body.
- It has two main parts:
  - ✓ **Blood** → circulates in blood vessels
  - ✓ **Interstitial fluid** → surrounds and lies between cells
- **Functions of ECF:**
  - ✓ Supplies **nutrients** to cells
  - ✓ Provides **ions and essential substances**
  - ✓ Helps in removal of **waste products**
- Normal body functioning depends on keeping the internal environment **constant (within physiological limits).** If the internal environment changes beyond limits, it leads to **malfunction or disease.**

## ❑ Role of Body Systems in Homeostasis

- Most body systems help in maintaining a **stable internal environment (homeostasis)**.
- Each system performs a **specific function** to keep the body balanced.

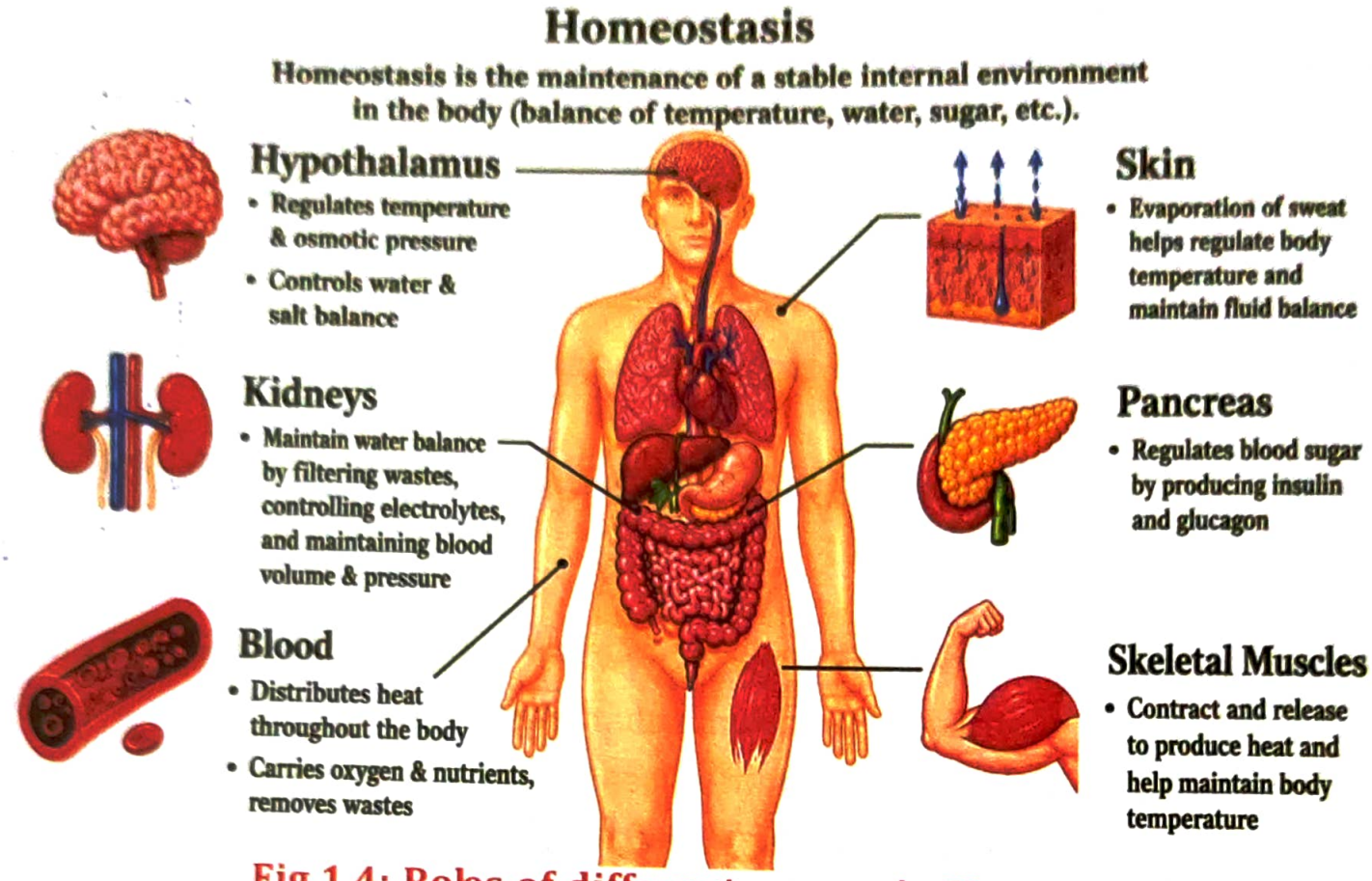



**Fig.1.4: Roles of different organs in Homeostasis**

## ❑ Role of body systems in homeostasis

### ❖ Digestive System

The digestive system provides nutrients required for growth and energy. It breaks down food into amino acids, carbohydrates, and fatty acids, absorbs these nutrients into the blood, and the blood transports them to all body cells.

### ❖ Respiratory System

The respiratory system helps in the exchange of gases by supplying oxygen for energy production and removing carbon dioxide through the lungs. It also helps maintain acid-base balance and body temperature.

### ❖ Excretory System

The excretory system removes waste products from the body. The kidneys filter the blood, eliminate toxic substances, and maintain water balance, electrolyte balance, and osmolarity.

### ❖ Integumentary System

The integumentary system acts as a protective barrier against injury, bacteria, and harmful substances. It protects the body from UV radiation and temperature changes, helps excrete small amounts of urea and salts, and detects touch, pain, and temperature.

### ❖ Blood and Circulatory System

The blood and circulatory system transports blood throughout the body, carrying oxygen, nutrients, and hormones to cells while removing carbon dioxide and metabolic wastes. White blood cells also help in defense and immunity.

### ❖ Lymphatic System

The lymphatic system carries lymph fluid through a network of vessels and returns proteins and excess fluid back to the blood. Lymph nodes destroy harmful substances and help maintain fluid balance.

### ❖ Endocrine System

The endocrine system secretes hormones that regulate metabolism, growth, and various body functions. It also helps maintain the body's internal balance.


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❖ **Skeletal Muscles**

Skeletal muscles help in body movement, protect internal organs from injury, and maintain body stability.

❖ **Nervous System**

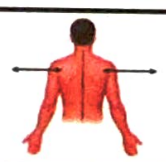
The nervous system controls and coordinates body activities. It detects changes in the environment, sends signals for quick responses, and regulates vital functions necessary for maintaining homeostasis.

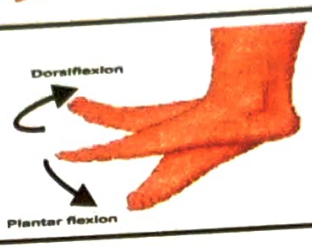
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## 1.1.6. Basic Anatomical Terminologies

❑ **Orientation and Directional Terms**

**Table 1.4: Orientation and Directional Terms**

| TERM | DEFINITION | ILLUSTRATION | EXAMPLE |
|---|---|---|---|
| **Superior (upper)** | Toward the head end or upper part of body; **above** |  | The forehead is superior to the nose. |
| **Inferior (lower)** | Away from the head end or toward the lower part of body; **below** |  | The stomach is inferior to lungs. |
| **Ventral (anterior)** | Toward or at the front of the body; **in front of** |  | The breastbone is anterior to the spine. |
| **Dorsal (posterior)** | Toward or at the backside of the body; **behind** |  | The heart is posterior to the sternum. |
| **Median (middle)** | Toward or at the midline of the body; **on the inner side of** |  | The heart is medial to the arm. |
| **Lateral (side)** | Away from the midline of the body; **on the outer side of** |  | The arms are lateral to the chest. |
| **Ipsilateral** | On the same side of the body as another structure |  | The gallbladder and ascending colon are ipsilateral. |
| **Contralateral** | On the opposite side of the body from another structure |  | The ascending and descending colons are contralateral. |
| **Intermediate** | Between a more medial and a more lateral structure |  | The collarbone is intermediate between breastbone and the shoulder. |
| **Proximal** | Close to the origin of the body part or the point of attachment of a limb to the body trunk |  | The elbow is proximal to the wrist. |
| **Distal** | Farther from the origin of a body part or the point of attachment of a limb to the body trunk |  | The knee is distal to the thigh. |

| | | | |
|---|---|---|---|
| **Superficial (external)** | Toward or at the body surface |  | The skin superficial to the skeleton. |
| **Deep (internal)** | Away from the body surface; **more internal** |  | The lungs are deep to the rib cage. |
| **Prone** | Lying flat, especially face downwards. |  | Yoga Cobra pose |
| **Supine** | Face up with their head resting on a pillow and their neck in a neutral position |  | Shavasana |
| **Palmar** | The front of the hand |  | |
| **Plantar** | Plantar flexion is a movement in which the top of foot points away from your leg. |  | |

## ❑ Anatomical Planes

- Imagine a person standing in an upright position. Now imagine dissecting this person with imaginary vertical and horizontal planes. This is the best way to describe anatomical planes.
- Anatomical planes are **hypothetical flat surfaces** used to **divide the body into sections**, allowing for precise descriptions of body locations, movements, and orientations and make it easy to describe or visualize the internal arrangement from different views.
- There are **four body planes that lie at right angles to each other**. These are:

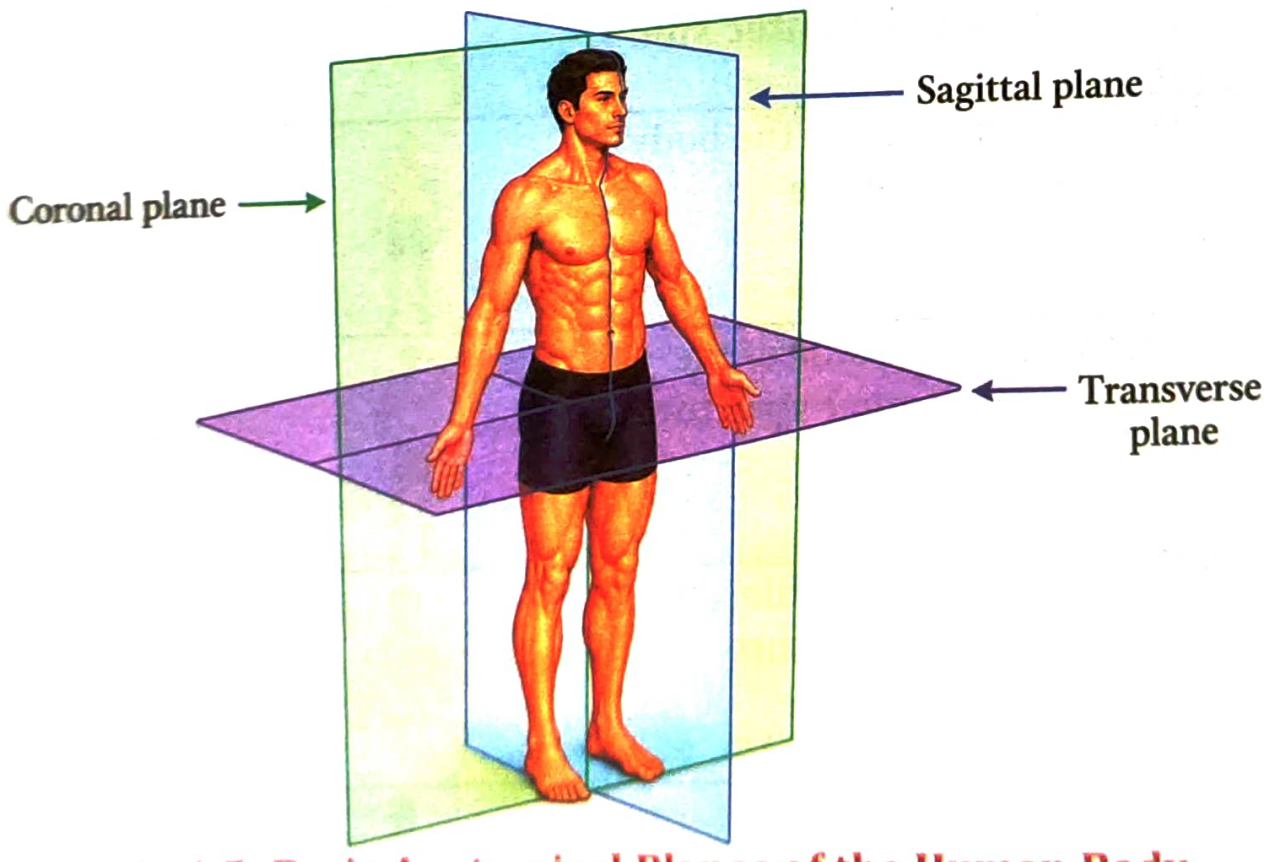



**Fig.1.5: Basic Anatomical Planes of the Human Body**

### ❖ Sagittal Plane

- The sagittal plane is a **vertical line that divides the body into right and left halves**.
- It allows movements like forward and backward movement.
- It is at a right angle (90°) to the frontal and transverse planes.
- The sagittal plane helps to study the position of organs in the **front (anterior) and back (posterior)** direction.


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➢ **Types of sagittal plane**

- ✓ **Midsagittal (Median) Plane:** Divides the body into equal left and right halves.
- ✓ **Parasagittal Plane:** Divides the body into unequal left and right parts.
  - **Examples:** Movements like walking or running occur in this plane.
  - The sagittal plane is the plane that runs parallel to the **middle or midsagittal plane.**
  - The viscera are shown anteroposteriorly on the sagittal plane.

❖ **Coronal Plane (Frontal Plane)**

- Coronal or frontal planes are those that **split the body into anterior (front) and posterior (back) sections** at right angle to the sagittal or median plane. It displays a viscera's anteroposterior extent.
- **Examples:** Movements like side lunges or lateral arm raises occur in this plane.

❖ **Transverse or Horizontal Plane**

- Imagine a horizontal plane running through the center of your body.
- This plane separates the body into two regions: **upper (superior) and lower (inferior).**

❖ **Oblique Plane**

- A two-dimensional, slanted plane that intersects the body in any direction other than parallel or perpendicular to the usual sagittal, coronal, or transverse planes.
- It passes through the body at **an angle other than 90 degrees.**

❑ **Anatomical movements**

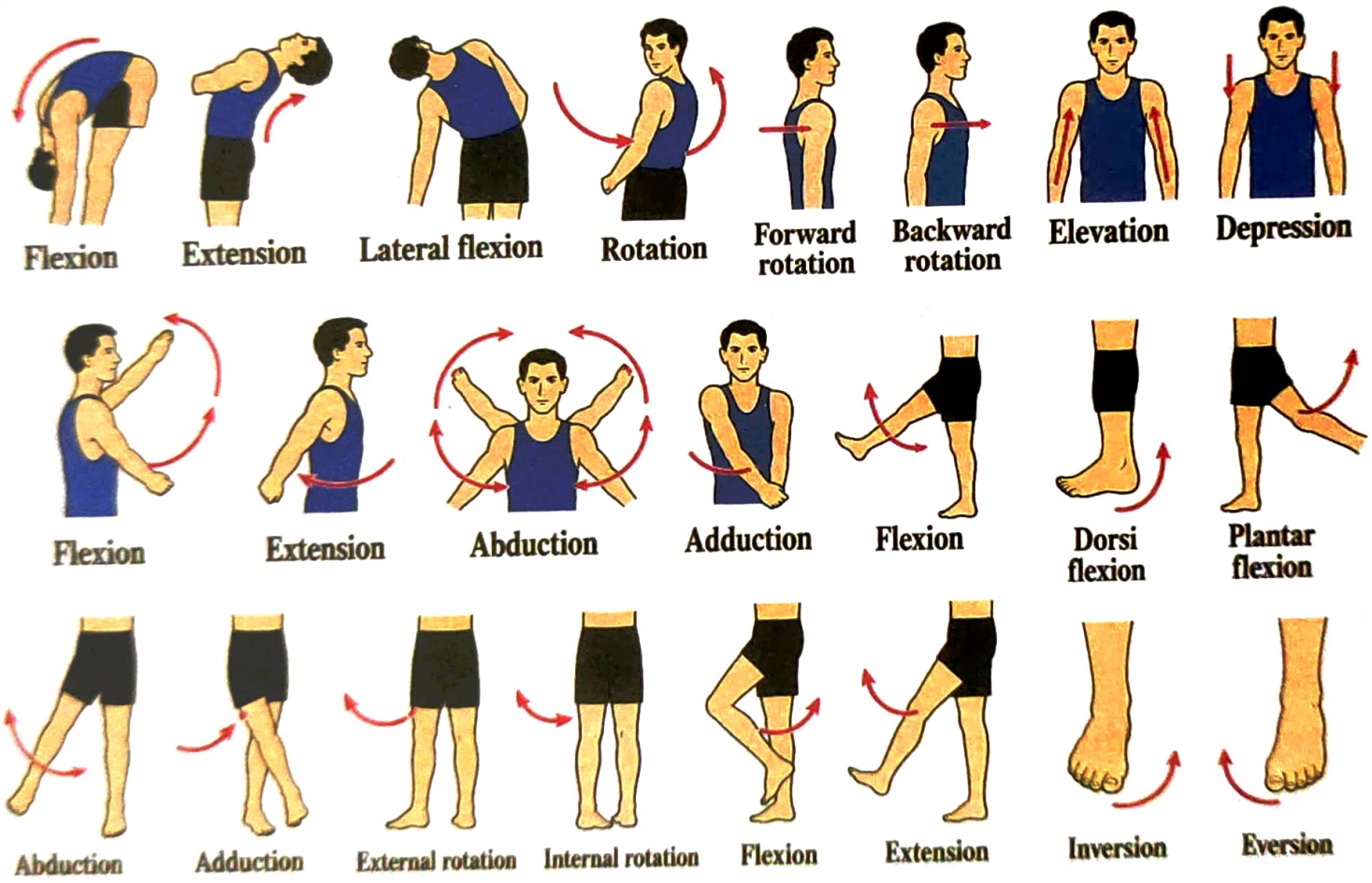



**Fig.1.6: Basic Anatomical Movements**

**Table 1.5: Basic Anatomical Movements**

| MOVEMENT | DEFINITION |
|---|---|
| **Flexion** | Bending |
| **Extension** | Straightening |
| **Abduction** | Moving away from the reference axis |
| **Adduction** | Bringing closer to the reference axis |
| **Medial Rotation** | Rotation toward the midline |
| **Lateral Rotation** | Rotation away from the midline |
| **Inversion** | Plantar side toward the medial plane |
| **Eversion** | Plantar side away from the medial plane |

| | |
|---|---|
| **Supination** | Lateral rotation of the radius, resulting in the palm of the hand facing anteriorly (if in anatomical position) or superiorly (if elbow is flexed). (**Mnemonic: Supinate to the Sun and Pronate to the Plants.** Supinate: palm towards the Sun; Pronate: palm towards the Plants) |
| **Pronation** | Medial rotation of the radius, resulting in the palm of the hand facing posteriorly (if in anatomical position) or inferiorly (if elbow is flexed) |
| **Protrusion** | Forward |
| **Retrusion** | Backward |
| **Elevation** | Superiorly to the reference axis |
| **Depression** | Inferiorly to the reference axis |
| **Circumduction** | Combination of flexion, abduction, extension, and adduction |
| **Deviation** | Ulnar and radial abduction of the wrist |
| **Opposition** | Bringing the thumb in contact to a finger |
| **Reposition** | Separating the thumb from the digits |
| **Plantar Flexion** | The movement of the foot in a downward motion away from the body |
| **Dorsal Flexion** | The movement that occurs when the foot and shin approximate closer together |

## 1.2. CELLULAR LEVEL OF ORGANIZATION

### ❑ Introduction to the Cell

All living things are composed of cells. Molecules combine to form cells. A cell is defined as the structural and functional unit of the living body. It can live, grow, and reproduce on its own. Cells are often called the building blocks of life. Each type of cell is made of chemicals and carries out specific chemical reactions.

### ❑ History of Cell

- **Robert Hooke** discovered the cell in **1665.** He gave the name of his discovery "Cells", which means "a small room" in Latin.
- Anton Van Leeuwenhoek first saw and described a live cell.
- In 1883, Robert Brown later discovered the Nucleus.

### ❑ Cell Theory

- Cell theory was jointly put forward by **Matthias Jacob Schleiden** and **Theodor Schwann** in 1839.
- Rudolf Virchow (1855) first explained that cells divide and new cells are formed from preexisting cells.

### ❑ Types of Cells

- Cells are broadly divided into two major types:
  - ✓ **Prokaryotic cells**
  - ✓ **Eukaryotic cells**

**Table 1.6: Comparison of Prokaryotic and Eukaryotic Cells**

| CHARACTERISTIC | PROKARYOTIC CELLS | EUKARYOTIC CELLS |
|---|---|---|
| **Complexity** | Simple in structure. | More complex in structure. |
| **Size** | Small (generally **1–5 µm**). | Large (generally **5–100 µm**). |
| **Organization** | Usually unicellular. | May be unicellular or multicellular. |
| **Cell Covering** | Cell is enclosed by a rigid **cell wall**. | Cell is enclosed by a flexible **plasma membrane**. |
| **Nucleus** | Lack a true membrane-bound nucleus. | Possess a well-defined nucleus. |
| **Subcellular Organelles** | Membrane-bound organelles are absent. | Distinct membrane-bound organelles are present (e.g., mitochondria, nucleus, lysosomes). |


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| | | |
|---|---|---|
| **Genetic Material** | DNA is located in the cytoplasm within the nucleoid region. | DNA is enclosed within the nucleus. |
| **Cell Division** | Usually occurs by **binary fission**; mitosis is absent. | Cell division occurs by **mitosis** (and meiosis in reproductive cells). |
| **Cytoplasm** | Membrane-bound organelles and cytoskeleton are absent. | Contains membrane-bound organelles and a cytoskeleton (a network of tubules and filaments). |
| **Ribosomes** | **70S ribosomes** are present. | **80S ribosomes** are present. |
| **Examples** | Bacteria and Archaea. | Animals, plants, fungi, and protists. |

## 1.2.1. Structure & Functions of The Cell:

- A cell consists of a **plasma membrane** enclosing a number of **organelles suspended in a watery fluid called cytosol.**
- The cell contents, excluding the nucleus, are made up of the cytoplasm, namely the **cytosol and other organelles.**

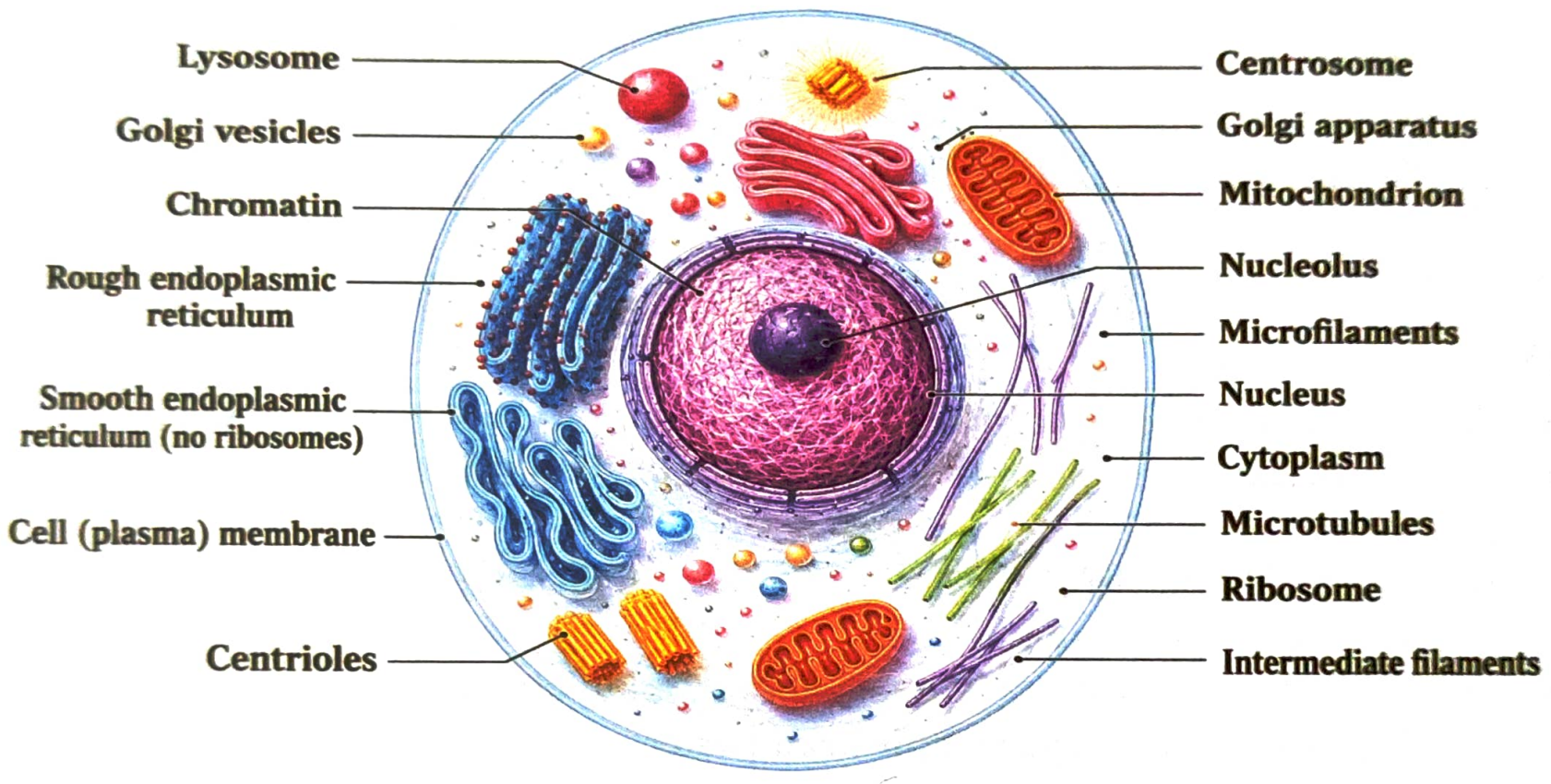



**Fig.1.7: Structure of a Typical Human Cell**

- Cell can be divided into three parts:
  - ✓ **Plasma Membrane**
  - ✓ **Cytoplasm**
  - ✓ **Nucleus**

### ❑ Plasma Membrane

- The plasma membrane also called **plasmalemma or cell membrane** is the outer covering of the cell.
- It is **selectively permeable,** which means it allows some substances to pass through while blocking others. It is flexible, responsive, and constantly changing in structure.
- This membrane separates one cell from another and helps maintain the internal environment of the cell by controlling the movement of ions and nutrients.
- It also has receptors that help in recognizing small molecules and plays an important role in interaction between cells.
- Under an electron microscope, the cell membrane appears about **7.2–8 nm thick.**
- It has a trilaminar (three-layered) structure, with proteins and sugars present within it.

- This three-layered structure is known as the **unit membrane.**
- It is made up of a **double layer (bimolecular layer) of lipid molecules,** which looks light under the microscope.
- This **lipid layer** is present in the middle and is surrounded on both sides by **two darker protein layers.**

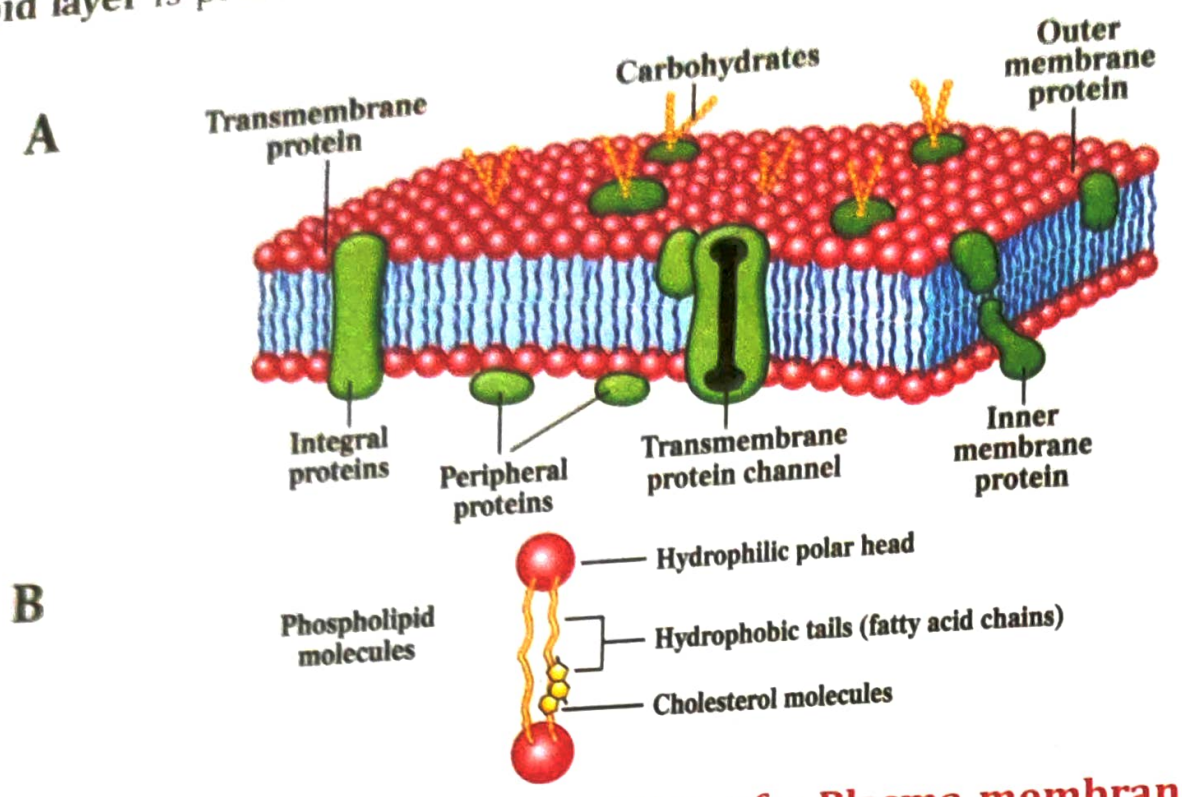



**Fig.1.8: Diagrammatic representation of a Plasma membrane**
**A. Fluid Mosaic Model of the Plasma Membrane**
**B. Structure of a Phospholipid Molecule**

### ❖ Fluid Mosaic Model

- The most accepted model of the cell membrane today is the **fluid mosaic model.**
- This model was further improved by **SJ Singer and GL Nicolson in 1972**, and is called the modified fluid mosaic model.
- According to this model, the cell membrane is flexible (fluid) and made up of different components arranged like a mosaic pattern.
- The lipid bilayer is made up of **phospholipid molecules**. Each molecule has a **hydrophobic tail** (water repelling) and a **hydrophilic head** (water-attracting).
- The hydrophilic heads face the outer and inner sides of the membrane, while the hydrophobic tails face towards the inside of the membrane.
- This arrangement helps to keep the inside and outside of the cell separate.

### ❖ Protein Layers of Cell Membrane

- The protein layers of the cell membrane are dense layers that cover both sides of the central lipid layer. These protein layers **protect the lipid layer.** The substances present in these layers are mainly glycoproteins.
- Membrane proteins run across the cell membrane and create pathways that help electrolytes and nutrients move in and out of the cell. These proteins have different roles:
  - ✓ Some work like enzymes.
  - ✓ Some act as receptors that attach to specific hormones or other substances.
  - ✓ Some form channels for the movement of small, water-soluble ions.
  - ✓ Their carbohydrate part helps the cell to be recognized (immunological identity).
  - ✓ They also help in the active transport of substances across the membrane.
- The carbohydrates of the membrane are present as **glycoproteins and glycolipids**. These carbohydrate parts extend outward from the cell surface. Both glycoproteins and glycolipids function as receptor sites.


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## ❑ Cytoplasm

- **Cytoplasm** is the part of the cell that surrounds the nucleus and is enclosed by the cell membrane. It **houses all organelles** (like mitochondria and ribosomes) except the nucleus, making it the site of metabolic activities such as protein synthesis.
- **Protoplasm comprises both the cytoplasm and the nucleus**. Cytoplasm is clear and gel-like in appearance. It mainly consists of water and contains organelles, enzymes, salts, and different organic substances. Organelles such as **ribosomes, mitochondria, endoplasmic reticulum, and vacuoles** are present in it.
- When observed under a light microscope, the cytoplasm is divided into two main parts:
  - ✓ **Cytoskeleton**
  - ✓ **Cytoplasmic organelles and inclusions**

### ❖ Cytoskeleton

- It is a **network of protein fibers**, that makes up the cytoskeleton.
- It supports the cell, keeps it in form, and aids in organelle organization and retention.
- It consists of:

#### ➢ Intermediate Filaments:

- Intermediate filaments are important components of the cell cytoskeleton that **provide mechanical strength and protect the cell from stress.**
- They help **maintain the position and stability** of organelles, especially the nucleus.
- They help form cell junctions and provide strength and support to tissues.

#### ➢ Microfilaments:

- Microfilaments are **very fine fibers made of actin** that help in **cell movement, maintain cell shape,** and are involved in **muscle contraction.**
- They are **contractile proteins measuring about 4–6 nm in diameter** and are mainly located at the outer part of the cell.
- These fibers provide support and help the cell change shape when needed.

#### ➢ Microtubules:

- Microtubules are **long, hollow, unbranched tubes** that act as tracks for the movement of mitochondria, chromosomes, and secretory granules within the cytoplasm.
- They are rigid structures that help **maintain cell shape and provide support to the cell.**

### ❖ Cytoplasmic Organelles

- Organelles mean **"small organs."** They have specific and specialized functions inside the cell and are usually surrounded by a membrane in the cytoplasm.
- Examples include the **nucleus, mitochondria, ribosomes, endoplasmic reticulum, Golgi apparatus, lysosomes, and cytoskeleton**.

#### ➢ Endoplasmic Reticulum

- The endoplasmic reticulum (ER) is composed of a network of vesicles (cisternae) and tubes (tubules). It is a **three-dimensional structure** and is surrounded by a **membrane about 80 Å thick.**
- All eukaryotic cells have an ER. In animal cells, the ER usually forms more than half of the total membrane content.
- There are two types of endoplasmic reticulum:

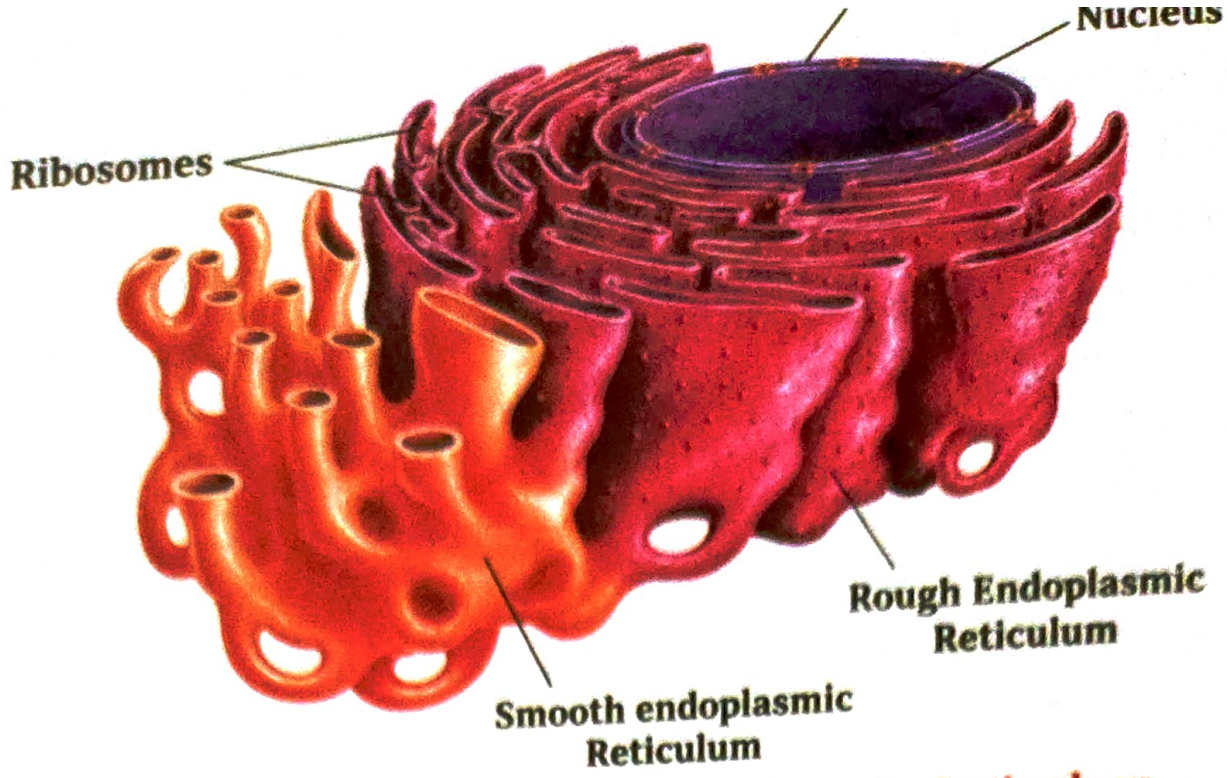



**Fig.1.9: Structure of Endoplasmic Reticulum**

✓ **Rough-Surfaced Endoplasmic Reticulum:**

- This type of reticulum contains osmophilic granules.
- **Ribosomes** are arranged in rows and **are attached to the membrane of the endoplasmic reticulum.**
- The rough appearance of the membrane is due to these granules.
- These **ribosomes** are the sites where proteins are synthesized.
- Some proteins, such as enzymes and hormones, may leave the cell and be used in other parts of the body.

✓ **Smooth-Surfaced Endoplasmic Reticulum:**

- There are no osmophilic granules in this type of endoplasmic reticulum.
- Ribosomes are present on the outer side of the membrane. Because of this, it appears smooth.
- The smooth endoplasmic reticulum carries out many metabolic processes.
- It produces lipids, phospholipids (for the plasma membrane), and steroids.
- These substances are secreted by cells of the testes, ovaries, and sebaceous glands.

✓ **Functions:**

- Protein synthesis takes place in the **rough endoplasmic reticulum**, which is found in large amounts in liver cells.
- Mammalian liver and gonad cells include the smooth endoplasmic reticulum, which is involved in making steroid hormones and may act as a detoxifying site in other cells.
- The sarcoplasmic reticulum in skeletal muscle plays a part in carrying impulses in the substance of muscle cells and is also involved in the binding of $Ca^{2+}$ ions.

➢ **Golgi Apparatus (Golgi Complex)**

- The **Golgi apparatus** was discovered by **Camillo Golgi in 1898.**
- It is a complex structure made of interconnected **tubules, vesicles, and cisternae.**
- It is present in most eukaryotic cells and is usually located near the nucleus and endoplasmic reticulum (ER).
- The Golgi apparatus is a membrane-bound organelle responsible for modifying, sorting, and packaging proteins and lipids for transport.
- It exhibits structural polarity with a cis (forming) face and a trans (maturing) face.
- Under an electron microscope, its main structures include:

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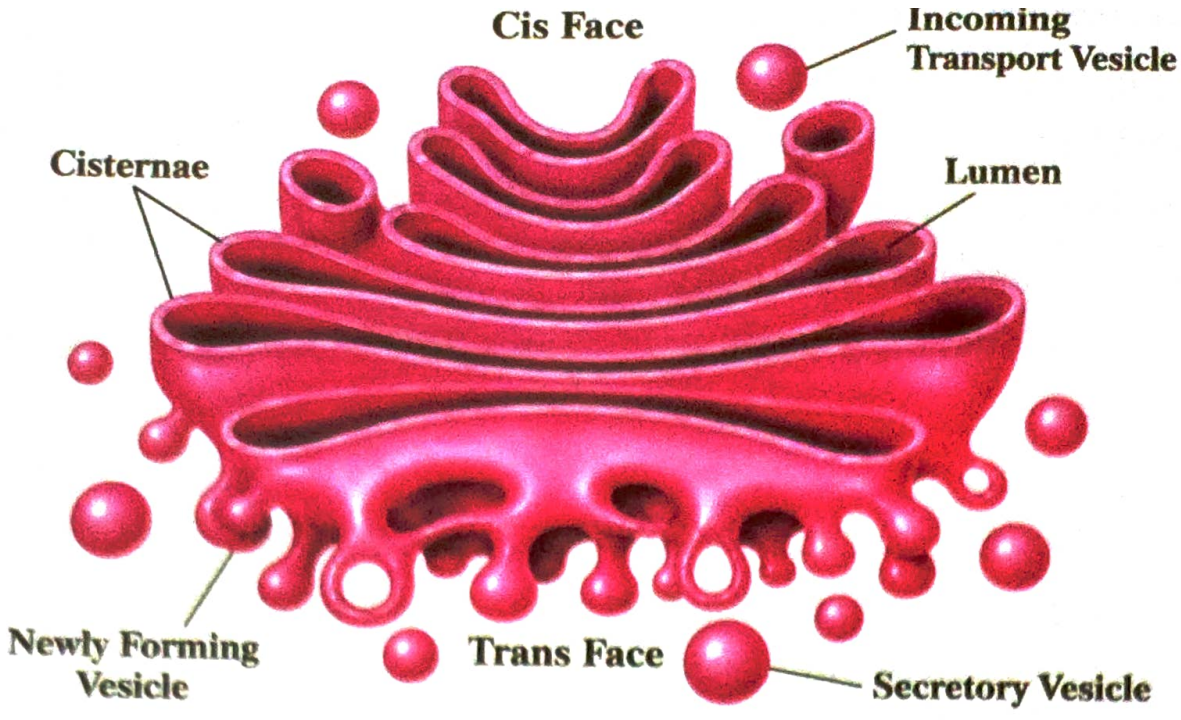



Fig.1.10: Structure of the Golgi Apparatus

✓ **Cisternae:**

- These are flattened sacs arranged in stacks. In longitudinal view, they appear like tubules.
- Each stack is called a dictyosome and usually contains 5–6 cisternae in animal cells.

✓ **Vesicles:**

- **Transitional vesicles:** Small membrane-bound vesicles formed from the ER. They move to the cis face of the Golgi and fuse to form new cisternae.
- **Secretory vesicles:** These vesicles become dense after filling with protein material. They release their contents outside the cell by exocytosis in response to signals.
- **Clathrin-coated vesicles:** These are small, spherical vesicles with a rough surface, found at the periphery of the Golgi. They are different from secretory vesicles and help in **transport of materials within the cell** (intracellular transport/pinocytosis).

✓ **Functions of Golgi Apparatus**

- It acts as a site for the **storage, processing, and packaging of cell products.**
- It modifies, concentrates, and transports substances received from the endoplasmic reticulum (ER).
- It helps in the formation of the plasma membrane, cell wall (in plants), and lysosomes.
- It **produces secretory granules and vacuoles** containing substances such as hormones, enzymes, and proteins.
- It synthesizes the polysaccharide part of glycoproteins.

➢ **Lysosomes**

- Lysosomes are small membrane-bound sacs (0.25–0.50 micron in size) **that contain digestive enzymes.**
- These enzymes break down large molecules like proteins, carbohydrates, lipids, and nucleic acids into smaller parts, which can then be used by the cell.
- Sometimes, **lysosomes can digest the cell's own contents**, so they are also called **"suicide bags."**

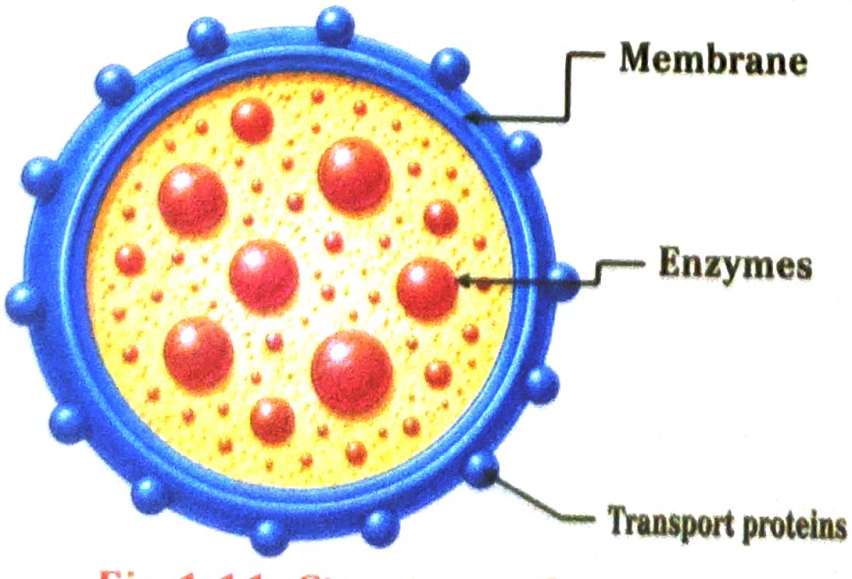



Fig.1.11: Structure of a Lysosome